UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 37605

LH EXOPLANETS

SUMMER EXAMINATIONS 2025

Time allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

Calculators may be used in this examination but must not be used to store text. Calculators with the ability to store text should have their memories deleted prior to the start of the examination.

A formula sheet and two tables of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

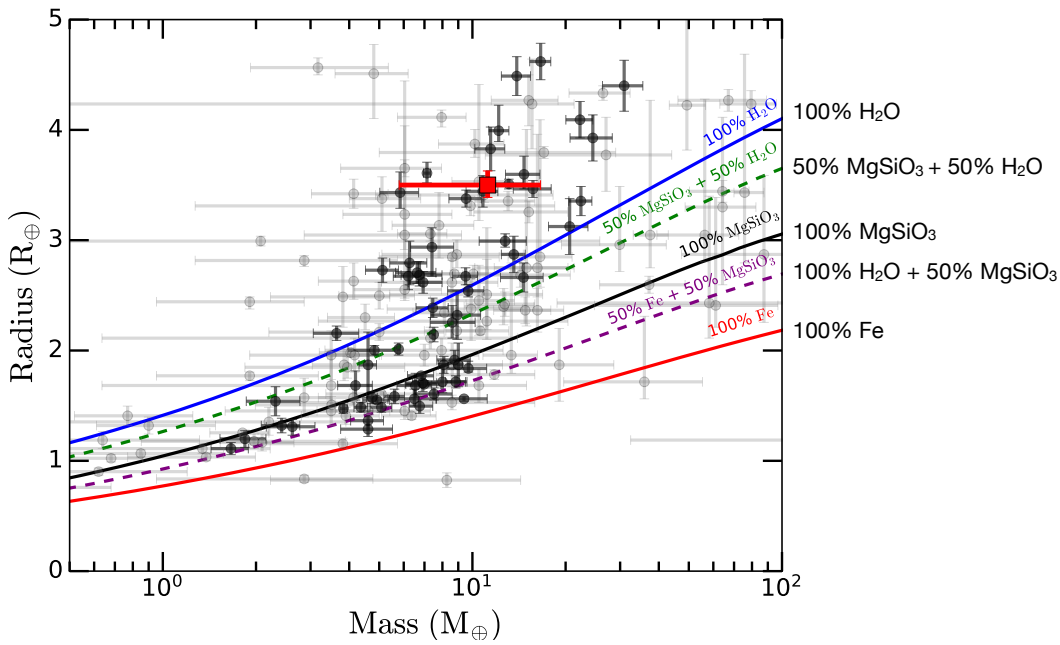
1. A $1.5\,\mathrm{R}_\oplus$ planet is detected in transit, but remains undetected with radial-velocities despite obtaining 9 measurements with a $30\,\mathrm{m\,s^{-1}}$ precision. Its orbital period is 0.12 day, and its host star is $0.5\mathrm{M}_\odot$ and $0.5\mathrm{R}_\odot$. This question is about estimating the planet's most likely mass using physics and knowledge from other exoplanetary systems. A mass/radius plot is provided with interior models. (Ignore the grey and red measurements from this plot, the least precise)

   (a) Starting from the semi-amplitude of a radial velocity signal, show that $K \sim 1\,\mathrm{m\,s^{-1}} \left(\frac{P}{1\ \mathrm{day}}\right)^{-1/3} \left(\frac{m_\mathrm{p}}{\mathrm{M}_\oplus}\right) \left(\frac{M_\star}{0.5\mathrm{M}_\odot}\right)^{-2/3}$. Then, calculate the maximum mass consistent with the current radial velocity measurements (in units of $\mathrm{M}_\oplus$). Assume a 3σ detection. State, and justify, any other assumptions. **[8]**

   (b) Starting from first principles, and using the formula sheet, derive an expression for the Roche radius. **[8]**

   (c) Using the expression for the Roche radius, estimate a minimum mass for the planet. Calculate this mass in units of $\mathrm{M}_\oplus$ and explain why this represents the minimum possible mass. **[6]**

   (d) Using the plot, estimate a refined mass for the planet, and provide a justified estimate for the uncertainty on that mass. Then calculate how many radial-velocity measurements are needed to make a more precise measurement than the estimate you just made. For that last part, assume the actual planet's mass $= 4\,\mathrm{M}_\oplus$. Provide a short comment about your answer. **[8]**



2. A protoplanetary dust disc is imaged and shows one dark lane, at $45\,\mathrm{AU}$ from its star. With an age of 10 Myr, the stellar mass, radius and effective temperature are $1.2\,\mathrm{M}_\odot, 1.3\,\mathrm{R}_\odot$ and 4800K respectively. JWST observations, using direct imaging at $6\,\mu\mathrm{m}$, then observe a dot of light at the same location as the lane, and measure its magnitude to be 7.1 fainter than the star, and its effective temperature to be $1200\,\mathrm{K}$.

   (a) Start with the equation of the disc scale height. First describe what each quantity is, then show that
   $H(a) \sim 0.21\,\mathrm{AU}\,\left(\frac{\mathrm{a}}{10\,\mathrm{AU}}\right)^{5/4}\left(\frac{\mathrm{T}_\star}{\mathrm{T}_\odot}\right)^{1/2}\left(\frac{\mathrm{M}_\odot}{\mathrm{M}_\star}\right)^{1/2}\left(\frac{18}{\mu}\right)^{1/2}\left(\frac{\mathrm{R}_\star}{\mathrm{R}_\odot}\right)^{1/4}$.
   Following that, use this expression to compute the scale-height at the location of the dark lane. State and justify any assumptions. **[8]**

   (b) Assuming an orbiting object is creating the lane in the disc, calculate its minimum mass. State your assumptions clearly, and justify why this is a minimum mass. What extra information would you need to calculate a maximum mass? **[7]**

   (c) Using Planck's law, estimate the radius of the object measured with direct imaging, within the lane. **[7]**

   (d) Describe the main steps that lead to the formation of a gas giant exoplanet with core accretion. Interpret the results you obtained in that light. At what stage is the system being studied here? What types of objects could this be, and how would one be able to verify? How will the radius and brightness of this object change as this system ages further? Be quantitative whenever you can. **[8]**

3. A periodic photometric event with a depth of 0.75% is observed on a Sun-like star. Its period is 11 days, and its duration is 4.3 hrs.

   (a) In the context of eclipsing binaries, and starting from eq. 4.1, show how to get to the primary and secondary eclipse depth (eq. 3.21). **[4]**

   (b) Assuming the object is a transiting planet, using eq. 3.21, find what values F_2 and $\mathcal{A}$ are (assume $F_1 = 100$). Using the observed values, calculate as much information about the system's possible stellar, planetary and orbital parameters as you can. What is needed to confirm this is a planet? State any assumptions you make, be quantitative, and report your results using astronomical units. **[7]**

   (c) There are at least four other scenarios able to create a periodic loss of flux of 0.75%. Describe three of them. Support your claim for each quantitatively (always take $F_1 = 100$, and give which possible values of F_2 and $\mathcal{A}$ you find). In addition, indicate how each scenario can be confirmed (or ruled-out) using other methods. Be quantitative here too. Each scenario has a range of possible ($F_1, F_2, \mathcal{A}$). **[15]**

   (d) In a few sentences, describe what limb darkening is, how it is produced, and what effect it produces on a planetary transit and/or a planetary secondary eclipse. **[4]**

Exoplanet Formula Sheet

1 Planet formation

1.1 Angular momentum: $L = m\,v\,a$

1.2 Keplerian velocity: $v_{\rm Kep} = \sqrt{\frac{G\,M_\star}{a}}$

1.3 Orbital Angular momentum: $L = m_{\rm p}\sqrt{GM_\star a}$

1.4 Virial Theorem: $2K + U = 0$

1.5 Gravitational potential (uniform sphere): $U_{\rm grav} = -\frac{3}{5}\,\frac{GM^2}{R}$

1.6 Kinetic energy of a gas particle: $K_{\rm gas} = \frac{3}{2}\,k\,T$

1.7 Jeans length: $R_{\rm jeans} = \sqrt{\frac{15\,k}{4\,\pi\,G}\,\frac{T}{\rho\,\mu\,m_{\rm p+}}}$

1.8 Jeans mass: $M_{\rm jeans} = \left(\frac{3}{4\,\pi\,\rho}\right)^{1/2}\left(\frac{5\,k\,T}{G\,\mu\,m_{\rm p+}}\right)^{3/2}$

1.9 Mass density: $\rho(z) = \mu\,m_{p^+}\,n(0)\,e^{-E(z)/kT(R)}$

1.10 Gravitational energy perpendicular to disc plane: $E(z) = \frac{G\,\mu\,m_{p^+}\,M_\star}{2R^3}\,z^2$

1.11 Protoplanetary disc scale height: $H = \sqrt{\frac{kT(a)a^3}{GM_\star\mu m_{\rm p+}}}$

1.12 Luminosity law: $L = 4\pi\,R^2\,\sigma\,T^4$

1.13 Equilibrium Temperature: $T_{\rm eq} = T_\star(1-A)^{1/4}\sqrt{\frac{R_\star}{2a}}$

1.14 Angular velocity: $\Omega = \frac{2\pi}{P} = \sqrt{\frac{G\,M_\star}{a^3}}$

1.15 The Hill sphere radius: $r_{\rm H} = a\sqrt[3]{\frac{m_{\rm p}}{3M_\star}}$

1.16 Escape velocity: $v_{\rm esc} = \sqrt{\frac{2Gm_{\rm p}}{r_{\rm p}}}$

1.17 Surface gravity: $g = \frac{Gm_{\rm p}}{r_{\rm p}^2}$

1.18 Mass accretion: $\frac{dm}{dt} = \pi\,r^2\,v\,\rho = \sigma\,v\,\rho$

1.19 Energy conservation: $\frac{1}{2}\,m'\,v_{\rm max}^2 - \frac{G\,m\,m'}{\Delta r} = \frac{1}{2}\,m'\,v^2$

1.20 Angular momentum conservation: $v_{\max}^2 = \frac{v^2\, b^2}{\Delta r^2}$

1.21 Safronov Number: $\Theta = \frac{v_{\rm esc}^2}{v^2}$

1.22 Mass accretion with gravitational focusing: $\frac{dm}{dt} = \pi r^2 v \rho \left(1 + \Theta\right)$

1.23 Sound speed (disc): $c_{\rm s} = \frac{h}{R} v_{\rm Kep}$

1.24 Observed mass accretion: $\frac{dm}{dt} = \frac{1}{2}\, \Sigma\, \Omega\, \pi\, r^2\, (1 + \Theta)$

1.25 Isolation mass is: $m_{\rm iso} = 8 \frac{\pi^{3/2}}{\sqrt{3}} a^3 M_\star^{-1/2} \Sigma^{3/2}$

1.26 Gas Accretion: $m \geq \sqrt{\frac{3}{32\,\pi}\, \frac{M_\star^3}{\rho\, a^3}}\, \left(\frac{H}{a}\right)^3$

1.27 Rotational energy: $R = \frac{1}{2}\, m\, \Omega^2\, r^2$

1.28 Toomre criterion: $Q = \frac{c_{\rm s}\, \Omega}{\pi\, G\, \Sigma} < 1$

1.29 Disc Collapse conditions: $\frac{m_{\rm d}}{M_\star} > \frac{h}{R}$

2 Two-body problem

2.1 Newton's law: $\boldsymbol{F}_1 = m_1\, \ddot{\boldsymbol{r}}_1 = +\frac{G m_1 m_2}{r^3} \boldsymbol{r}$

2.2 Newton's law: $\boldsymbol{F}_2 = m_2\, \ddot{\boldsymbol{r}}_2 = -\frac{G m_1 m_2}{r^3} \boldsymbol{r}$

2.3 Equation of Motion: $\ddot{\mathbf{r}} + \mu\, \frac{\mathbf{r}}{r^3} = 0$

2.4 Mass Scale: $\mu = G\, (m_1 + m_2)$

2.5 Planar motion: $\mathbf{r} \times \dot{\mathbf{r}} = \mathbf{h}$

2.6 Polar coordinates: $\mathbf{r} = r\, \hat{\mathbf{r}}$

2.7 Polar coordinates: $\dot{\mathbf{r}} = \dot{r}\, \hat{\mathbf{r}} + r\, \dot{\theta}\, \hat{\theta}$

2.8 Polar coordinates: $\ddot{\mathbf{r}} = \left(\ddot{r} - r\, \dot{\theta}^2\right) \hat{\mathbf{r}} + \left(\frac{1}{r}\, \frac{d}{dt}\left(r^2 \dot{\theta}\right)\right) \hat{\theta}$

2.9 Angular momentum: $\mathbf{h} = r^2\, \dot{\theta}\, \hat{\mathbf{z}}$

2.10 Conic's equation: $r = \frac{h^2/\mu}{1 + e\, \cos(\theta - \varpi)}$

2.11 Specific angular momentum: $h = \sqrt{\mu a (1 - e^2)}$

2.12 Distance to the star: $r = \frac{a\, (1 - e^2)}{1 + e\, \cos f}$

2.13 Kepler's second law: $\dot{A} = \frac{1}{2}\sqrt{\mu a(1-e^2)}$

2.14 Kepler's third law: $P^2 = \frac{4\pi^2}{\mu} a^3$

2.15 Periastron velocity: $v_{\rm peri}^2 = \frac{G\,M_\star}{a}\,\frac{(1+e)}{(1-e)}$

2.16 Vis-Viva equation: $v_{\rm pl}^2 = G\,M_\star\,\left(\frac{2}{r} - \frac{1}{a}\right)$

2.17 Area of an ellipse: $A = \pi a b = \pi a^2\sqrt{1-e^2}$

2.18 Orbital angles relations: $\varpi = \Omega + \omega; \theta = f + \varpi$

3 Indirect detection methods

3.1 Doppler effect: $\lambda_{\rm obs} = \lambda_0\,\frac{1+\frac{v}{c}}{\left(1-\frac{v^2}{c^2}\right)^{1/2}}$

3.2 Radial velocity: $V_{\rm rad} = v\sin i = \frac{\Delta\lambda}{\lambda_0}\,c$

3.3 R: $\mathbf{R} = \frac{m_1\,\mathbf{r}_1 + m_2\,\mathbf{r}_2}{m_1+m_2}$

3.4 $\ddot{R}$: $\ddot{\mathbf{R}} = \frac{m_1\,\ddot{\mathbf{r}}_1 + m_2\,\ddot{\mathbf{r}}_2}{m_1+m_2} = 0$

3.5 Reflex motion: $R_2 = \frac{m_1}{M_{\rm tot}}\,a = a_2$

3.6 Reflex motion: $R_1 = \frac{m_2}{M_{\rm tot}}\,a = a_1$

3.7 Radial velocity: $V_{\rm rad} = \dot{z} = \sin i\left[\dot{r}\,\sin(f+\omega) + r\,\dot{f}\,\cos(f+\omega)\right]$

3.8 $\dot{r}$: $\dot{r} = \frac{r\,\dot{f}\,e\,\sin f}{1+e\,\cos f}$

3.9 Radial velocity equation: $V_{\rm rad} = \frac{2\pi a}{P}\frac{\sin i}{\sqrt{1-e^2}}[\cos(f+\omega) + e\cos\omega] + \gamma$

3.10 Semi-amplitude: $K = \frac{2\,\pi\,a}{P}\frac{\sin i}{\sqrt{1-e^2}}$

3.11 Motion caused by a planet: $K_\star = \frac{m_{\rm p}}{M_\star + m_{\rm p}}\frac{2\pi}{P}\frac{a\sin i}{\sqrt{1-e^2}}$

3.12 Motion caused by a star: $K_{\rm p} = \frac{M_\star}{M_\star + m_{\rm p}}\frac{2\pi}{P}\frac{a\sin i}{\sqrt{1-e^2}}$

3.13 Planet's mass function: $f(m_{\rm p}) = \frac{m_{\rm p}^3\sin^3 i}{(M_\star+m_{\rm p})^2} = (1-e^2)^{3/2}\frac{PK_\star^3}{2\pi G}$

3.14 Transit probability: $\mathcal{P} \sim \frac{R_\star}{a}$

3.15 Transit depth: $D = \frac{r_{\rm p}^2}{R_\star^2}$

3.16 Scaled radius: $\frac{R_\star}{a} = \frac{W}{P} \frac{\pi}{\sqrt{(1+\sqrt{D})^2 - b^2}} \frac{1+e\sin\omega}{\sqrt{1-e^2}}$

3.17 Distance from stellar centre: $\mu = \sqrt{1-(X^2+Y^2)}$

3.18 Limb darkening law: $I = 1 - u_a(1-\mu) - u_b(1-\mu)^2$

3.19 Stellar density: $\frac{M_\star}{R_\star^3} \sim \left(\frac{a}{R_\star}\right)^3 \frac{4\pi^2}{GP^2}$

3.20 Planet gravity: $g_{\rm p} = \frac{2\pi}{P} \frac{K_\star\sqrt{1-e^2}}{\sin i} \left(\frac{a}{r_{\rm p}}\right)^2$

3.21 Eclipse Depths: $D_{\rm pri} = \frac{(F_1+F_2)-\left(\mathcal{A}_{\rm pri}F_1+F_2\right)}{F_1+F_2}$ $\quad D_{\rm sec} = \frac{(F_1+F_2)-(\mathcal{A}_{\rm sec}F_2+F_1)}{F_1+F_2}$

3.22 Eclipse duration ratio: $\frac{W_{\rm occ}}{W_{\rm tra}} \sim \frac{1+e\sin\omega}{1-e\sin\omega}$

3.23 Eclipse time: $\Delta t \sim \frac{T}{2}\left(1 + \frac{4}{\pi}\, e\cos\omega\right)$

3.24 Projected separation: $\delta(f) = \frac{1-e^2}{1+e\cos f}\sqrt{1-\sin^2 i \sin^2(f+\omega)}$

3.25 Eclipse criterion: $\delta(f) < \frac{R_1+R_2}{a}$

3.26 Transit Ephemeris: $T_n = T_0 + n \times T$

3.27 Right Ascension: $\alpha^\star = \alpha\cos\delta$

3.28 Parallax: $\varpi = 1'' \left(\frac{1\ \rm pc}{d}\right)$

3.29 Astrometric displacement: $\Lambda = 1\,\rm arcsec \left(\frac{m_{\rm p}}{M_\star}\right) \left(\frac{a}{1\,\rm AU}\right) \left(\frac{d}{1\,\rm pc}\right)^{-1}$

3.30 Light deflection angle: $\epsilon = \frac{4GM_{\rm L}}{rc^2}$

3.31 Lens equation: $\theta_1^2 - \alpha\,\theta_1 - \theta_{\rm E}^2 = 0$

3.32 Einstein angle: $\theta_{\rm E} = \sqrt{\frac{4GM_{\rm L}}{\mu D_{\rm L} c^2}}$ with $\mu = 1 + \frac{D_{\rm L}}{D_{\rm LS}}$

3.33 Einstein radius: $r_{\rm E} = \theta_{\rm E}\, D_{\rm L}$

3.34 Image position: $\theta_\pm = \frac{\theta_{\rm E}}{2}\left(u \pm \sqrt{u^2+4}\right)$

3.35 Image position: $\theta_\pm = \frac{1}{2}\alpha \pm \left[\left(\frac{1}{2}\alpha\right)^2 + \theta_{\rm E}^2\right]^{1/2}$

3.36 Lens magnification: $A_\pm = \left|\frac{\theta_\pm}{\alpha}\frac{d\theta_\pm}{d\alpha}\right|$

3.37 Microlensing amplitude: $A = \frac{u^2+2}{u\sqrt{u^2+4}}$

3.38 Duration of microlensing: $t_{\rm E} = \theta_{\rm E}/\nu$

4 Atmospheres

4.1 Eclipse Depth: $D(\lambda) = \frac{F_{\rm out} - F_{\rm in}}{F_{\rm out}}$

4.2 Transmission spectroscopy: $D(\lambda) = D + D'(\lambda)$

4.3 Transmission spectroscopy: $D(\lambda) = D + \frac{2 r_{\rm p} H A_\lambda}{R_\star^2}$

4.4 Atmospheric scale height: $H = \frac{kT}{\mu m_{\rm p+} g}$

4.5 Thermal velocity: $v_{\rm therm} = \sqrt{\frac{3kT}{\mu m_{p+}}}$

4.6 Emission flux: $\Delta F_{\rm em}(\lambda) = D \frac{B_\lambda(T_{\rm p})}{B_\lambda(T_\star)} \sim D \frac{T_{\rm p}}{T_\star}$

4.7 Reflected flux: $\Delta F_{\rm ref}(\lambda) = A_\lambda \left(\frac{r_{\rm p}}{a}\right)^2$

4.8 Occultation depth: $\Delta F_{\rm occ}(\lambda) = \Delta F_{\rm em}(\lambda) + \Delta F_{\rm ref}(\lambda)$

4.9 Apparent magnitude: $m_1 - m_2 = -2.5 \log_{10} \left(\frac{F_1}{F_2}\right) = -2.5 \log_{10} \Delta F_{\rm occ}$

4.10 Absolute magnitude: $m_1 - M_1 = 5 \log_{10} d/10$

4.11 Signal to Noise ratio: $SNR \sim \frac{A}{\sigma_i/\sqrt{N}}$

4.12 Inner edge of the habitable zone: $a_{\rm HZ,in} = a_\oplus \times \sqrt{\frac{1}{2}}$

4.13 Outer edge of the habitable zone: $a_{\rm HZ,out} s = a_\oplus \times \sqrt{2}$

5 Know thy star

5.1 Planck's law: $B_\lambda(T) = \frac{2hc^2}{\lambda^5} \frac{1}{e^{hc/(\lambda k T)} - 1}$

5.2 Wien's law: $\lambda_{\rm top} = \frac{\kappa}{T_{\rm eff}}$ where $\kappa = 0.0028978\,{\rm m\,K}$

5.3 Sound speed (atmosphere): $c_{\rm s}^2 = \frac{\partial P}{\partial \rho}$

5.4 Ideal gas law: $P = \frac{n \mathcal{R} T}{V}$

5.5 Sound speed (atmosphere): $c_{\rm s}^2 = \gamma \frac{\mathcal{R} T}{\mathcal{M}}$

5.6 Pressure (inside a star): $P \propto \frac{G M_\star^2}{R_\star^4}$

5.7 Large separation scaling relation: $\Delta\nu = \Delta\nu_\odot \left(\frac{M_\star}{M_\odot}\right)^{1/2} \left(\frac{R_\star}{R_\odot}\right)^{-3/2}$

5.8 Acoustic cut-off frequency: $\nu_{\rm ac} = \frac{c_{\rm s}}{2\,H_{\rm p}}$

5.9 Pressure scale-height: $H_{\rm p} = \frac{k\,T}{\mu\,m_{p+}\,g}$

5.10 Maximum frequency scaling relation: $\nu_{\rm max} = \nu_{\rm max,\odot}\left(\frac{M_\star}{M_\odot}\right)\left(\frac{R_\star}{R_\odot}\right)^{-2}\left(\frac{T_{\rm eff}}{T_{\rm eff,\odot}}\right)^{-1/2}$

5.11 Scaling relation: $V_{\rm osc} = 23.4\left(\frac{L_\star}{L_\odot}\right)\left(\frac{M_\star}{M_\odot}\right)^{-1}\,{\rm cm\,s^{-1}}$

5.12 Scaling relation: $\frac{\delta L_\star}{L_\star} = 4.7\left(\frac{L_\star}{L_\odot}\right)\left(\frac{M_\star}{M_\odot}\right)^{-1}\left(\frac{T_{\rm eff}}{T_\odot}\right)^{-2}\left(\frac{\lambda}{550\ {\rm nm}}\right)^{-1}\,{\rm ppm}$

5.13 Radial-velocity precision: $\sigma_{\rm RV} \sim \frac{\sqrt{\rm FWHM}}{C\times{\rm SNR}}$

5.14 Skumanich law: $\Omega_\star = \frac{v_\star}{R_\star} \propto t^{-1/2}$

6 Non-Keplerian effects & Migration

6.1 Type I migration timescale: $\tau_{\rm I} = \frac{1}{2}\frac{H/a}{m_{\rm p}/M_\star}\frac{M_\star}{\Sigma a^2}\Omega_{\rm p}^{-1}$

6.2 Type I migration timescale: $\tau_e \sim 2.6(H/a)^2\tau_{\rm I}$ and $\tau_i \sim 3.7(H/a)^2\tau_{\rm I}$

6.3 Type II migration timescale: $\tau_{\rm II} = \frac{a^2}{\alpha H^2\Omega}\times\max\left(1, \frac{m_{\rm p}}{4\pi\Sigma a^2}\right)$

6.4 Planet separation: $\kappa = \frac{a_2-a_1}{\Delta r_{\rm H}}$

6.5 Effective planet separation: $\kappa_{\rm eff} = \kappa - 0.5\sigma_\kappa$

6.6 Mutual Hill radii: $\Delta r_{\rm H} = \frac{a_1+a_2}{2}\left(\frac{m_{\rm p,1}+m_{\rm p,2}}{3M_\star}\right)^{1/3}$

6.7 Tidal circularisation semi-major axis: $a_f = a_p\,(1+e_i)$

6.8 Tidal acceleration: $a_{\rm tides} = \pm\frac{2GM_\star\Delta r}{r^3}$

6.9 Roche radius: $a_{\rm Roche} = r_{\rm p}\sqrt[3]{\frac{3\,M_\star}{m_{\rm p}}}$

6.10 mass parcel: $\delta m = \frac{1}{2}\,k_2\,m_{\rm p}\left(\frac{r_{\rm p}}{a_{\rm peri}}\right)^3$

6.11 change in orbital energy: $\dot{E} = \frac{dE}{dt} = \frac{2\pi\,\Delta E}{Q\,T}$

6.12 energy to deform the planet: $\Delta E = \frac{1}{2}\,\Delta I\,\omega^2$

6.13 condition for pseudo synchronisation: $\omega = \frac{v_{\rm peri}}{a_{\rm peri}}$

6.14 Inertia for a spherical body: $I_{\rm sph} = \frac{2}{5}\, m_{\rm p}\, r_{\rm p}^2$

6.15 Inertia for a dumbbell: $I_{\rm dum} = 2\, \delta m\, r_{\rm p}^2$

6.16 Change in inertia: $\Delta I = \frac{6}{5}\, \delta m\, r_{\rm p}^2$

6.17 Change in orbital energy: $\Delta E = \frac{3}{5}\, k_2 \left(\frac{r_{\rm p}}{a_{\rm peri}}\right)^5 \frac{(1+e)}{(1-e)}\, E$

6.18 Tidal timescale: $\frac{1}{\tau_{\rm tides}} = \frac{\dot{E}}{E}$

6.19 Tidal timescale: $\tau_{\rm tides} = \frac{5}{6\pi} \frac{Q\,T}{k_2} \left(\frac{a_{\rm peri}}{r_{\rm p}}\right)^5 \frac{1-e}{1+e} \propto \left(\frac{a_{\rm peri}}{r_{\rm p}}\right)^5$

6.20 Tidal circularisation timescale: $\tau_{\rm circ} = \frac{4}{63} Q \sqrt{\frac{a^3}{G M_\star}} \left(\frac{m_{\rm p}}{M_\star}\right) \left(\frac{a}{r_{\rm p}}\right)^5$

6.21 Tidal synchronisation timescale: $\tau_{\rm syn} = Q \left(\frac{r_{\rm p}^3}{G m_{\rm p}}\right) \left(\frac{m_{\rm p}}{M_\star}\right)^2 \left(\frac{a}{r_{\rm p}}\right)^6 (\omega_{\rm p} - \omega_{\rm syn})$

6.22 Tidal infall timescale: $\tau_{\rm infall} \propto \left(\frac{a}{R_\star}\right)^5 \left(\frac{M_\star}{m_{\rm p}}\right)$

6.23 Rossiter-McLaughlin effect amplitude: $K_{\rm RM} = \frac{2}{3} D\, v \sin i_\star \sqrt{1 - b^2}$

6.24 Spin-orbit angle: $\cos\psi = \sin i_\star \cos\lambda \sin i_{\rm p} + \cos i_\star \cos i_{\rm p}$

Astrophysical Constants and Units

Astronomical Unit	AU	1.50×10^{11} m
Parsec	pc	3.1×10^{16} m
Tropical year	y	365.242 mean solar days
Solar luminosity	$L_\odot$	3.84×10^{26} W
Absolute bolometric magnitude of the Sun	$M_{\rm bol}$	$+4^{\rm m}75$
Bolometric correction for the Sun	BC	$-0^{\rm m}08$
Apparent visual magnitude of the Sun	$m_v(\odot)$	$-26^{\rm m}74$
Solar constant	f	1.36×10^{3} W m^{-2}
Solar mass	$M_\odot$	1.989×10^{30} kg
Solar radius	$R_\odot$	6.95×10^{8} m
Solar effective temperature	$T_{\rm eff}$	5777 K
Wien constant	b	2.898×10^{-3} m K
Hubble constant	H_o	70 km s^{-1} Mpc^{-1}
Distance of the Sun from galactic centre		8.3 kpc
Earth mass	$M_\oplus$	5.972×10^{24} kg
Earth radius	$R_\oplus$	6371 km
Jupiter mass	$M_{\rm Jup}$	1.898×10^{27} kg
Jupiter radius	$R_{\rm Jup}$	71 492 km

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.